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Network Layer: IPv4 Functions

Lecture 9 | CSE421 – Computer Networks

Department of Computer Science and Engineering
School of Data & Science

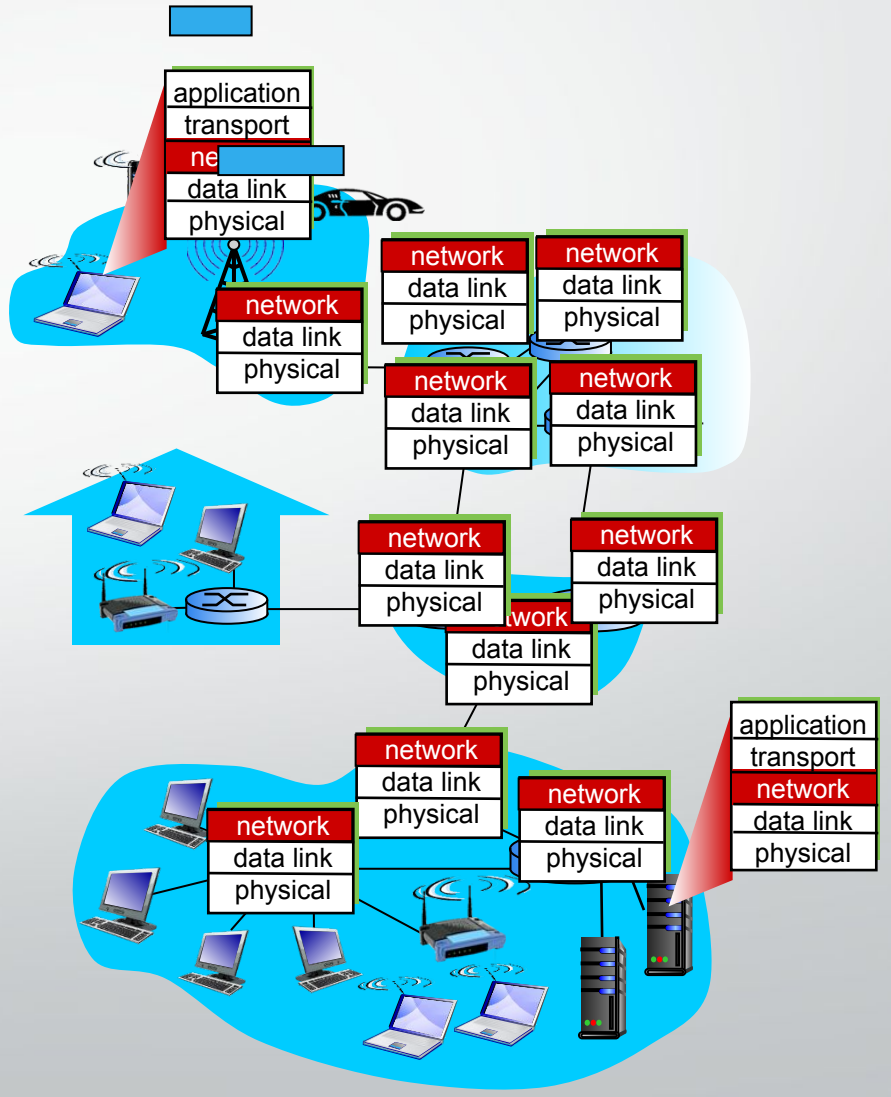
Objectives

- Short overview of the Network Layer
- Packet Switching: Virtual Circuits & Datagram Networks
- IPv4 Packet Format
- IP Fragmentation & Reassembly
- ICMP
  - Ping
  - Traceroute

The Network Layer

- Encapsulates data into **packets** on the sending side.
- Network Layer protocols operates on **hosts** and **routers**.
- **Routers** inspect IP header fields for forwarding.

Delivers segments to the **transport layer** on the



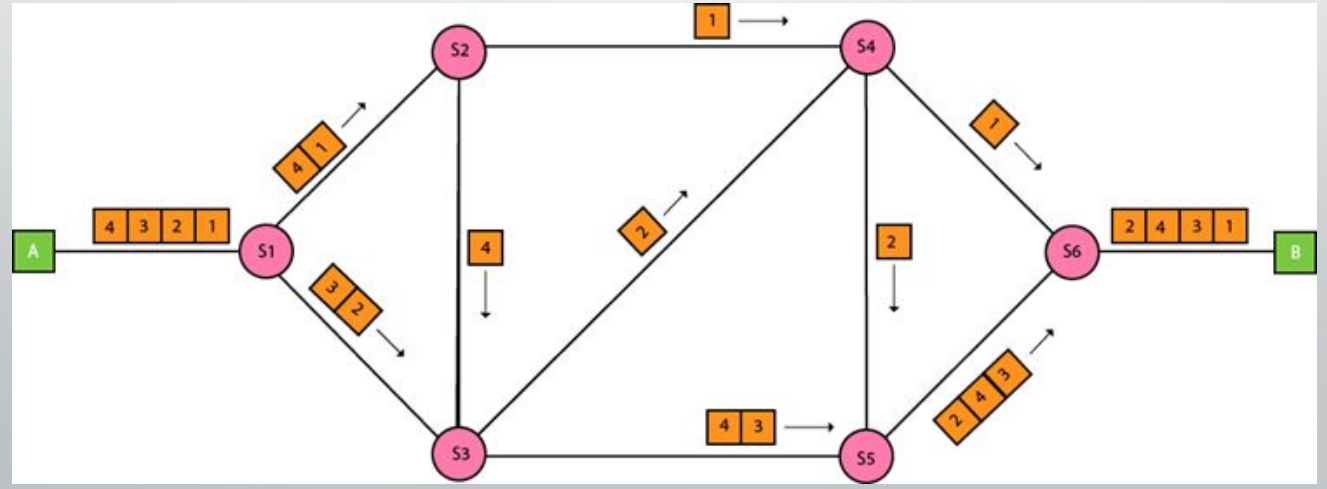
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Packet Switching

Packet Switching

- Packet Switching is a method of transferring data across a network by breaking it into smaller packets.

- Two type of networks based on packet switching
  - **Datagram Networks**
  - **Virtual Circuit Networks**



Datagram networks

- **No Call Setup:**

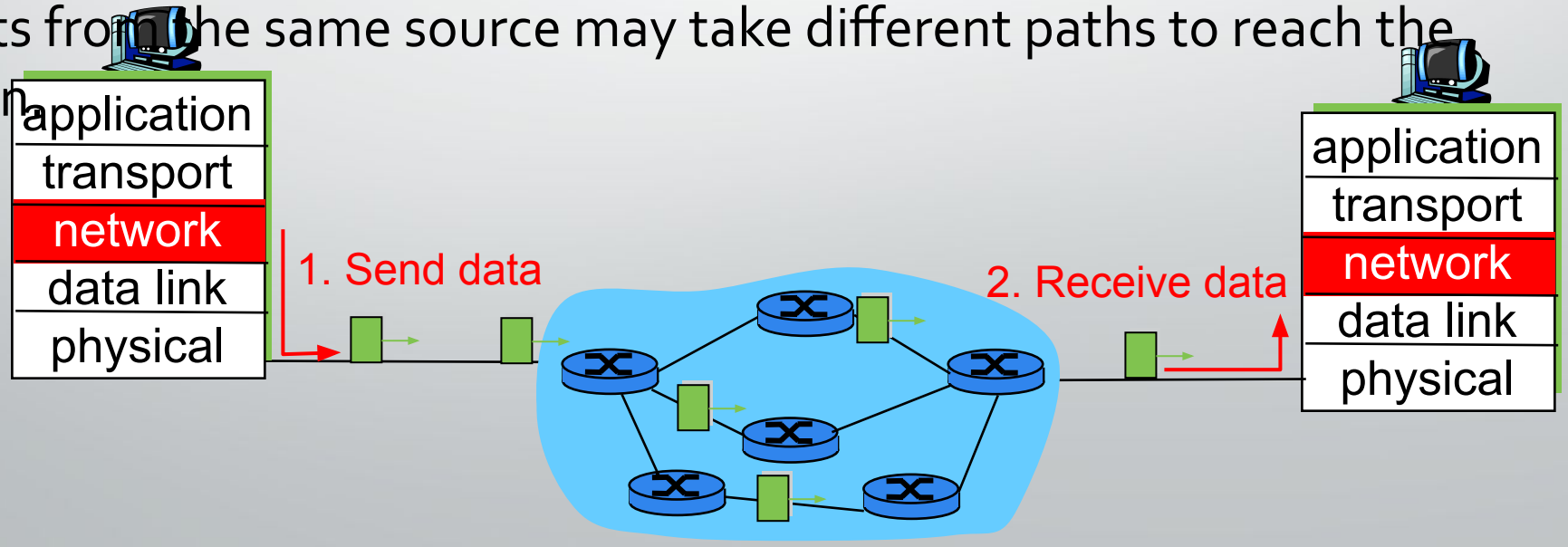
  Devices can send data immediately without establishing a connection.

- **Stateless Routers:**

  Routers forward packets independently based on their destination IP address.

- **Packet Forwarding:**

  Packets from the same source may take different paths to reach the destination.



Virtual Circuits: Signaling Protocols

- **Call Setup:**

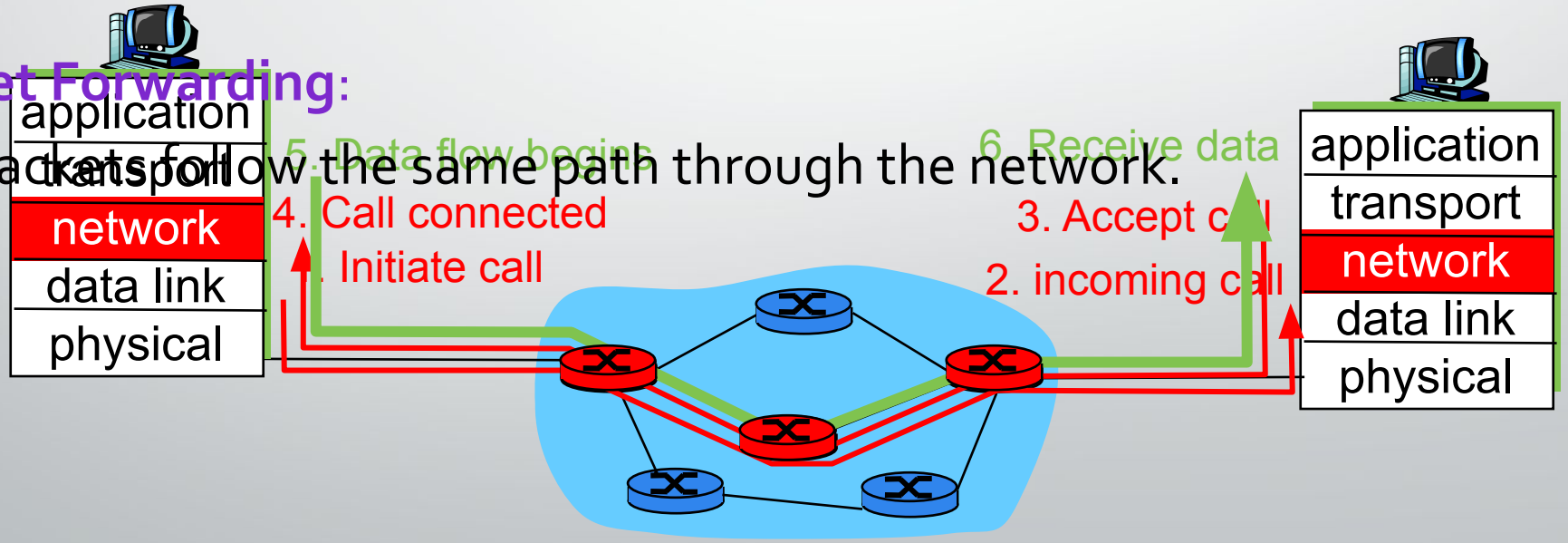
  A connection (virtual circuit) is established between sender and receiver before data transfer.

- **Stateful Routers:**

  Routers maintain information about active connections (virtual circuits).

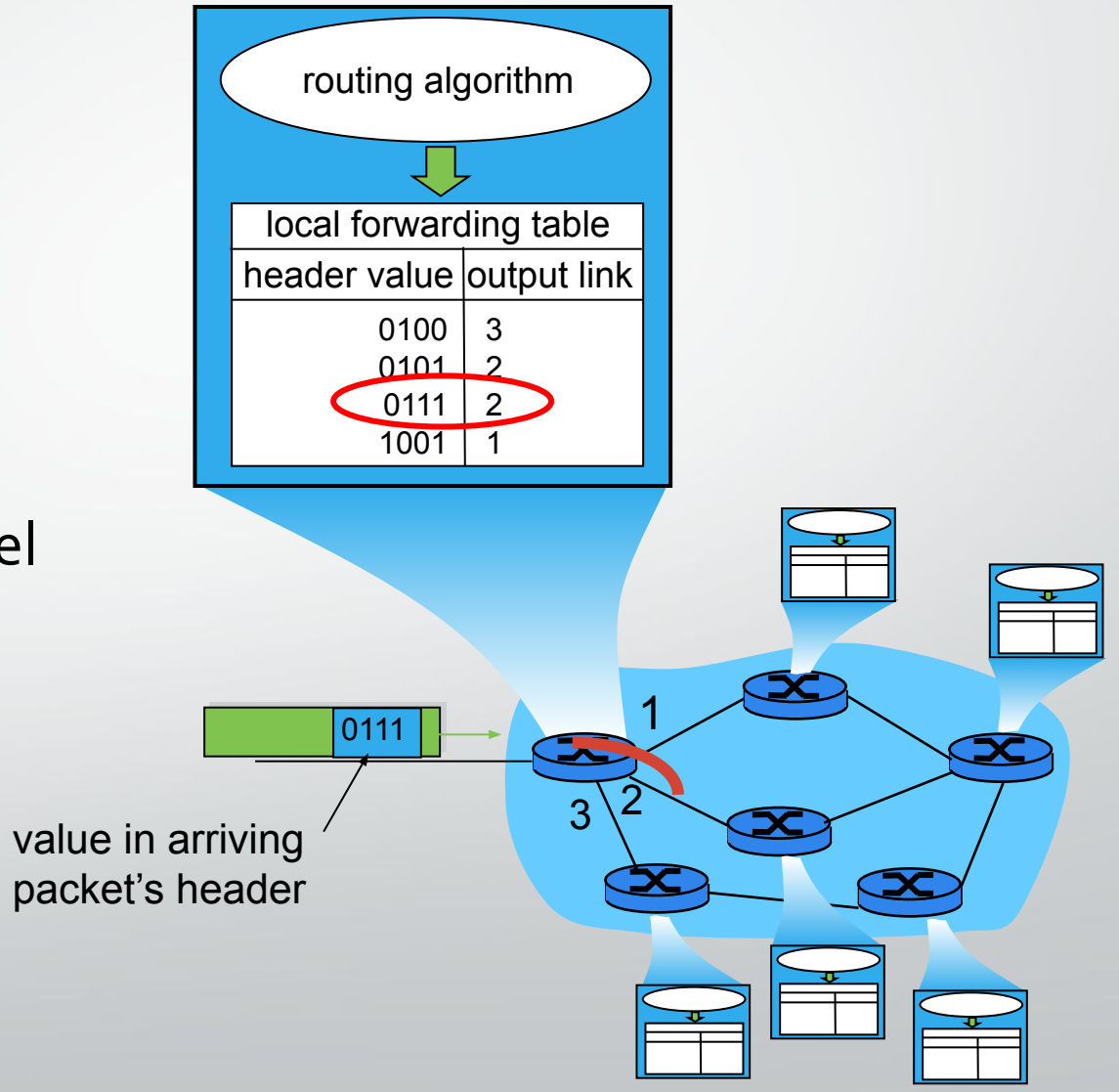
- **Packet Forwarding:**

  All packets follow the same path through the network.



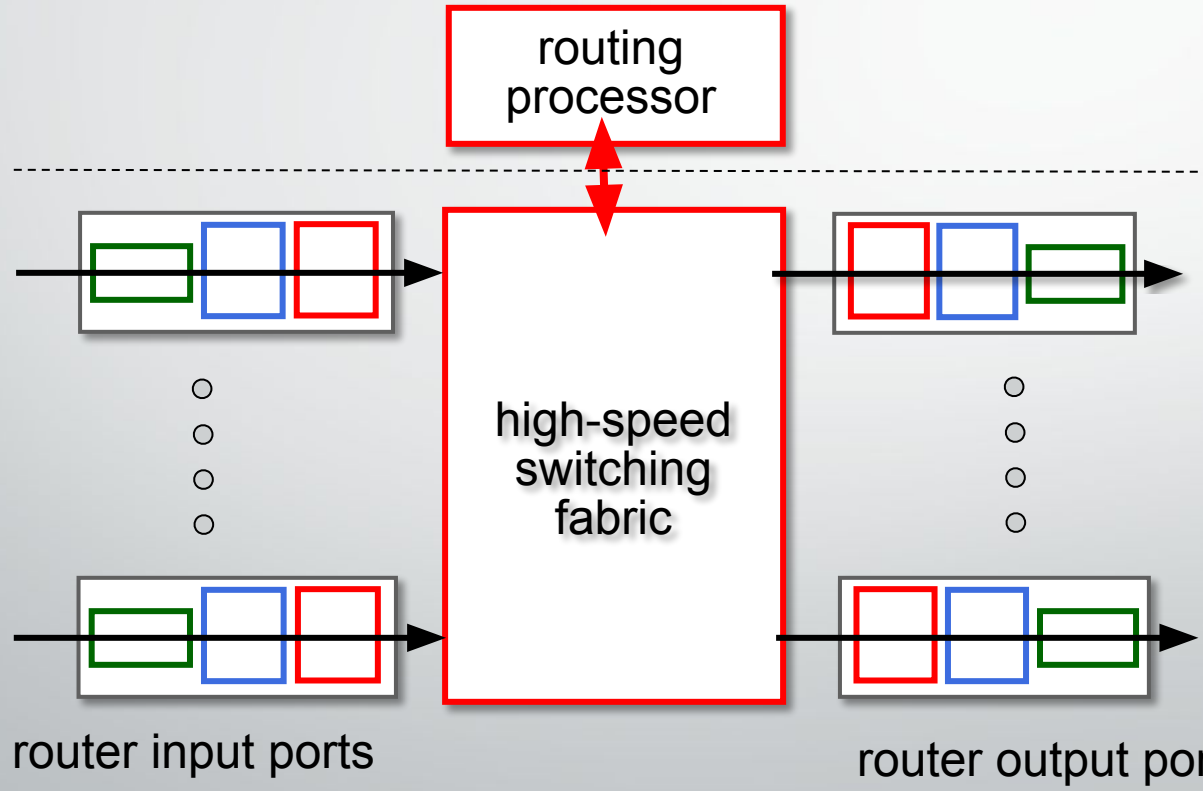
Functions of Network Layer

- **Routing:**
  - Finds the best path from source → destination
  - Done by routing algorithms
  - Analogy: planning a trip before you travel
- **Forwarding:**
  - move packets from router's input to appropriate router output
  - Analogy: process of getting through a single interchange



Router architecture overview

high-level view of generic router architecture:



routing, management control plane **(software)** operates in millisecond time frame



forwarding data plane **(hardware)** operates in nanosecond timeframe



Control Plane – the brain!

- Control plane controls the network using routing protocols.
- It creates the **forwarding table FIB** (Forwarding Information Base), a simplified version of routing table.
- The FIB may be created using
  - simple destination-based rules
  - or more advanced generalized rules.

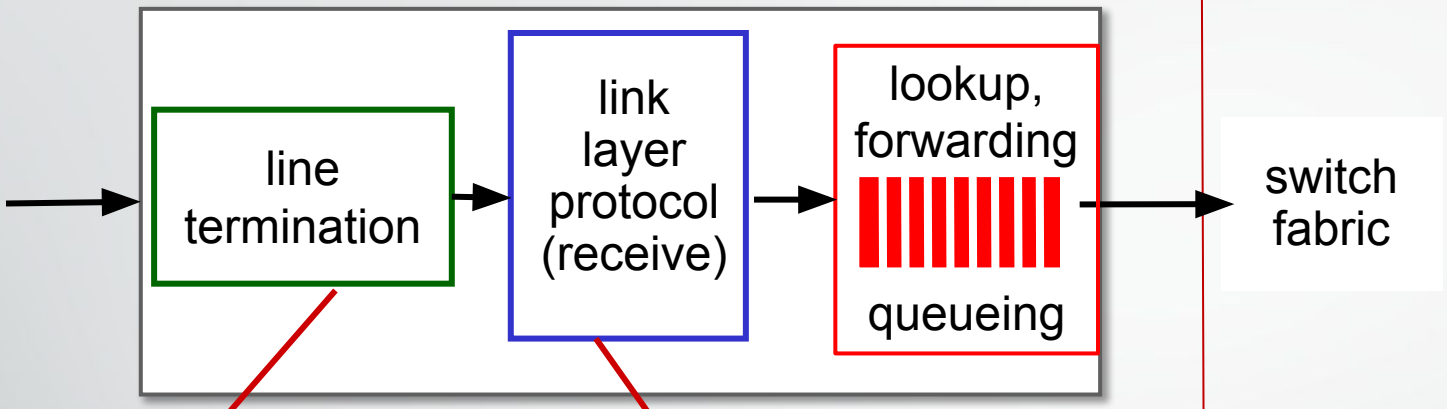
Control plane installs this FIB into every input port.

Decentralized Switching

Data Plane – the hands and legs!

- How input ports use the FIB?
- Forwarding rules can be:
  - **Destination-based forwarding :**
    - Uses **only destination IP** to choose output port
    - If destination = 200.20.20.0/24 → send to port s0/0/0.
  - **Generalized forwarding :**
    - Admins can add **multiple header fields** (IP, protocol, port, etc.) Or special rules (QoS)
    - If video packet (UDP, port 5000) → high-priority queue.
- These rules are stored in the FIB created by the control

Input port functions



physical layer:

- Converts signal → digital bits

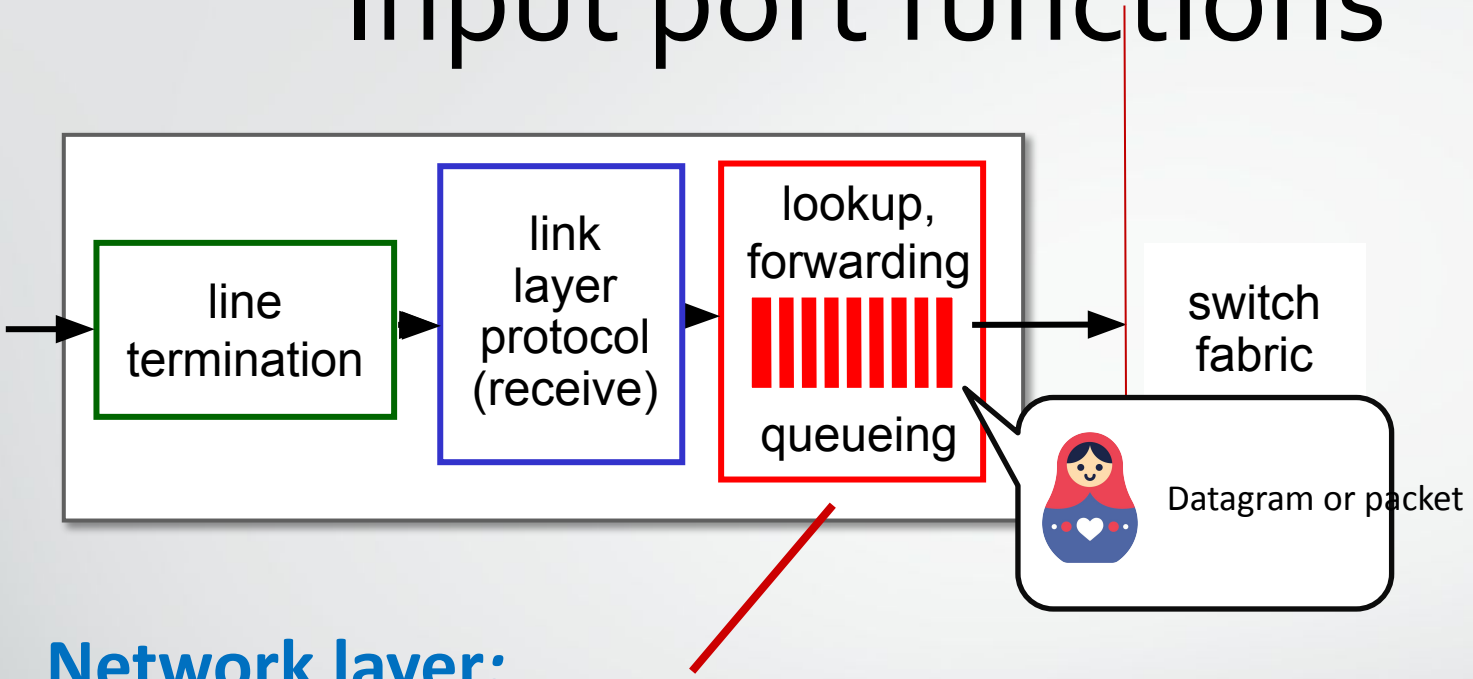
link layer:

- Removes data link header/trailer
- Checks for errors
- Extracts the IP packet



Frame

Input port functions



Network layer:

- Input port checks packet header
- Matches against local FIB copy
- Decides output port (using destination-based or generalized rule)

Input Port Queueing

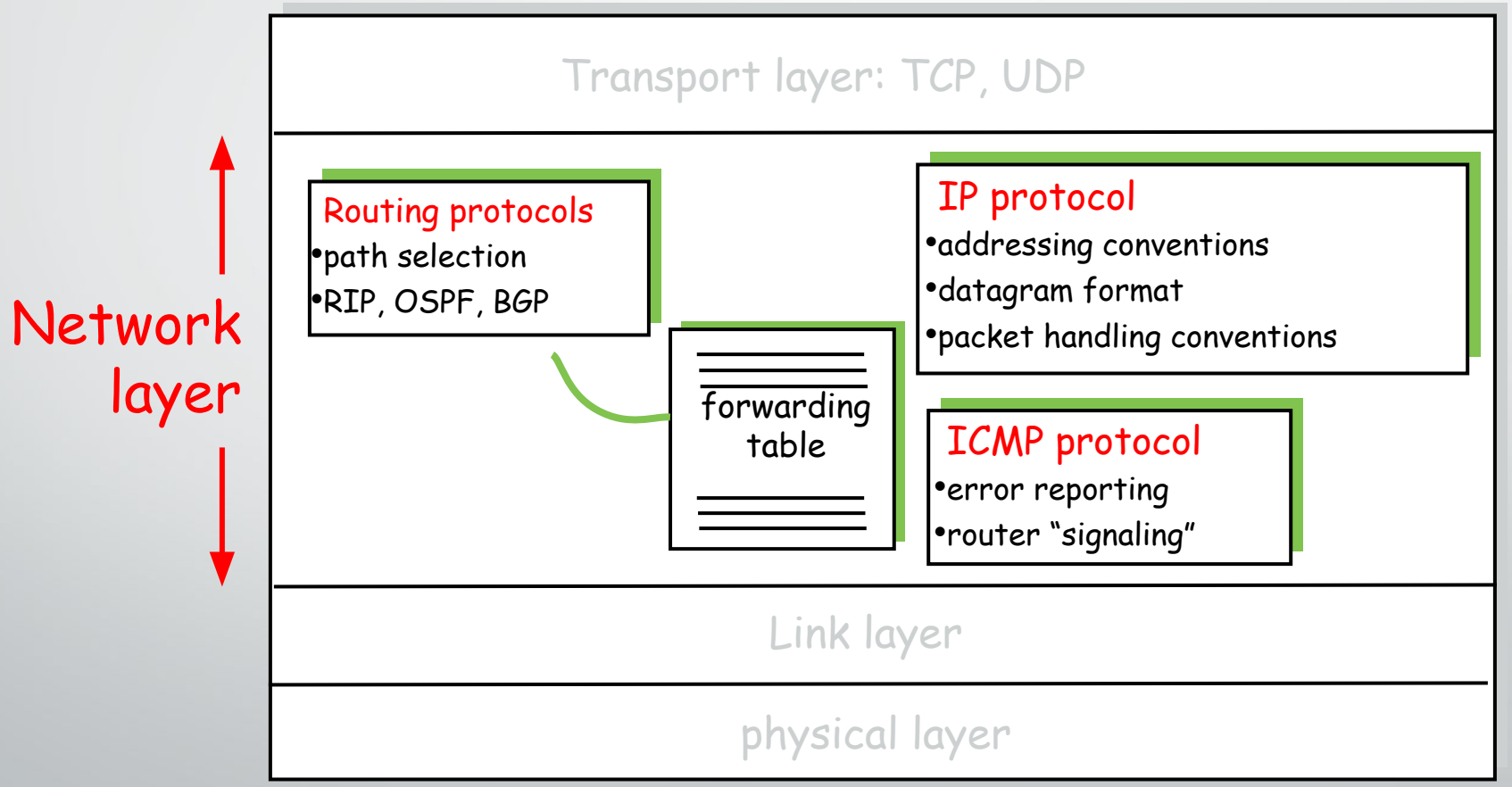
- If packets arrive too fast → they wait here
- If queue is full → packet is dropped

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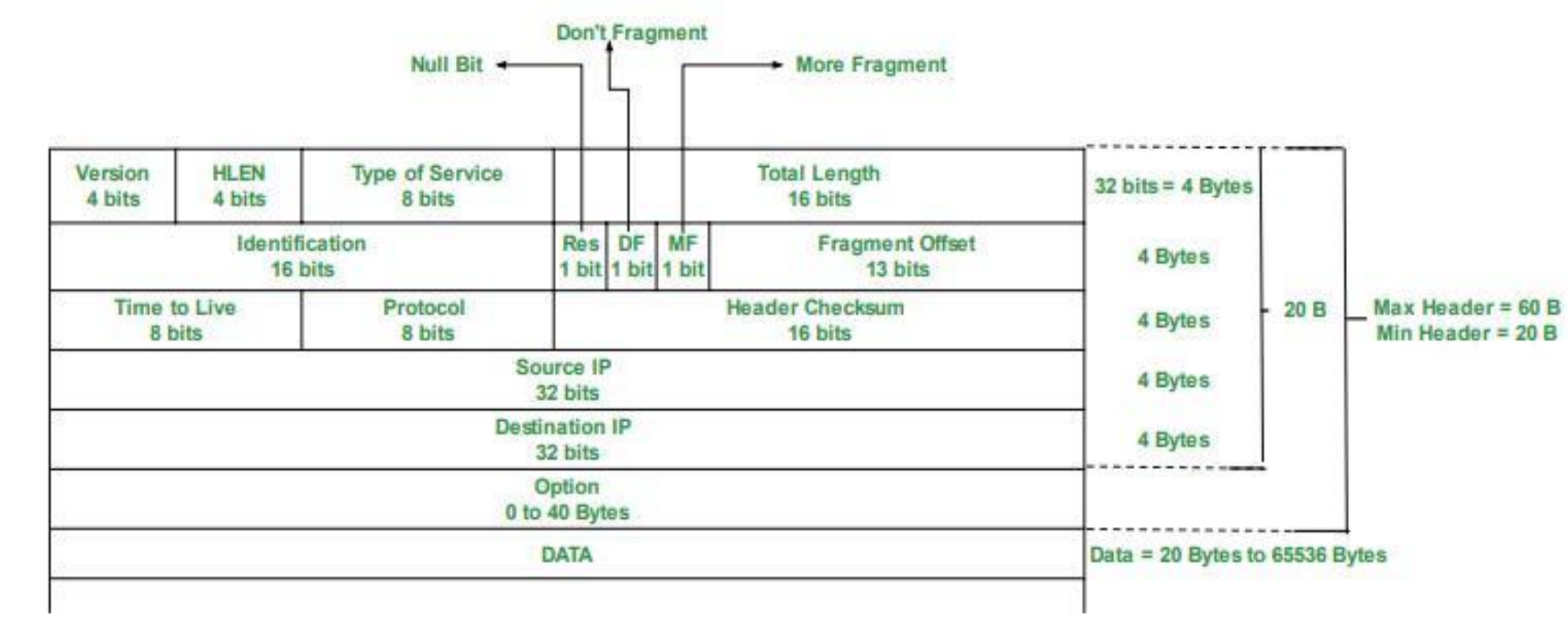
Internet Protocol IPv4

Internet Network Layer

- Host, router network layer functions:



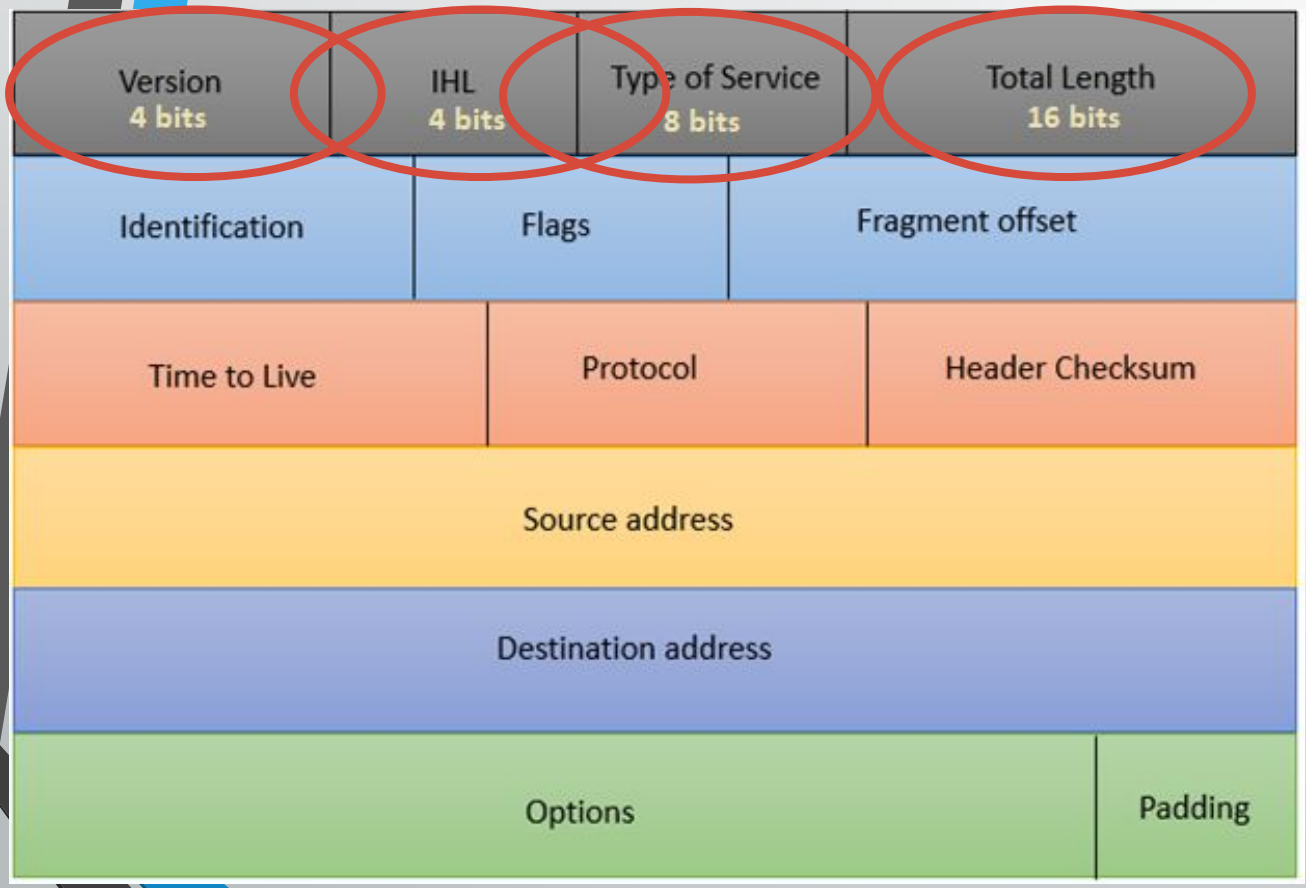
IPv4 Datagram Format



The size of an IP datagram:

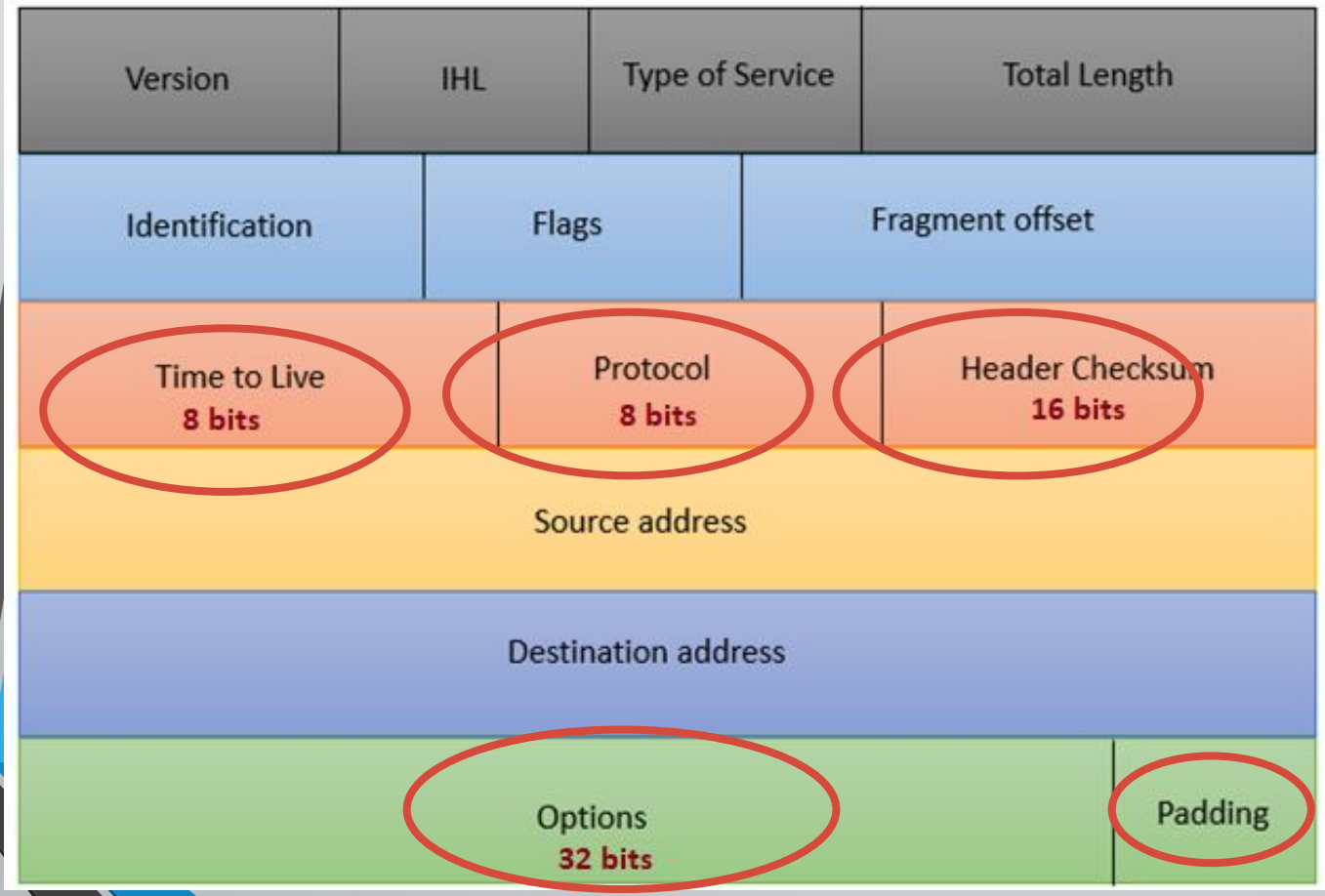
- The **minimum** size is **20 bytes** (if you have no data)
- The **maximum** size is **65,535 bytes**

IPv4 Datagram Header Format



- **Version:** value of which IP version is being used. For IPv4 the value will be 4 here.
- **Internet Header Length**: value of the header length, min 20 bytes, max 60 bytes. Shown in 4 byte word. **So min value 5, max 15.**
- **Type of Service**: for QoS (Quality of Service). To mark the packet to give special treatment or priority.
- **Total Length:** value of the entire size of the IP packet (header and data) in bytes.

IPv4 Datagram Header Format

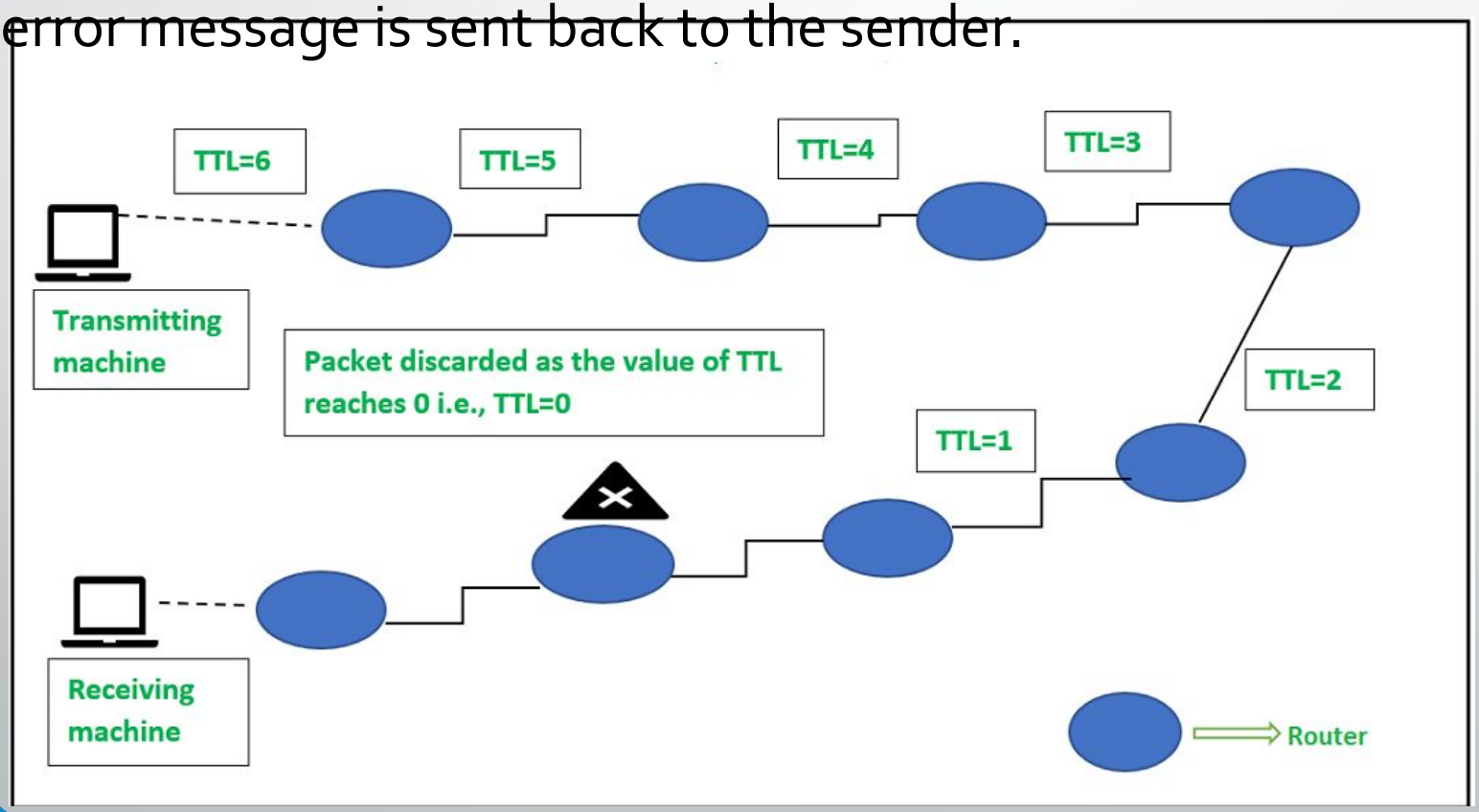


- **Time to Live:** maximum number of **hops** (routers) a packet can travel
- **Protocol:** value tells us which upper layer protocol is present, for example **TCP** has value **6** and **UDP** has value **17**.
- **Header Checksum:** to check if there are any errors in the header.
- **Options:** value of any extra information. Options are rarely used now.
- **Padding:**

Used only when Options are present. Ensures the header length becomes a multiple of 4 bytes

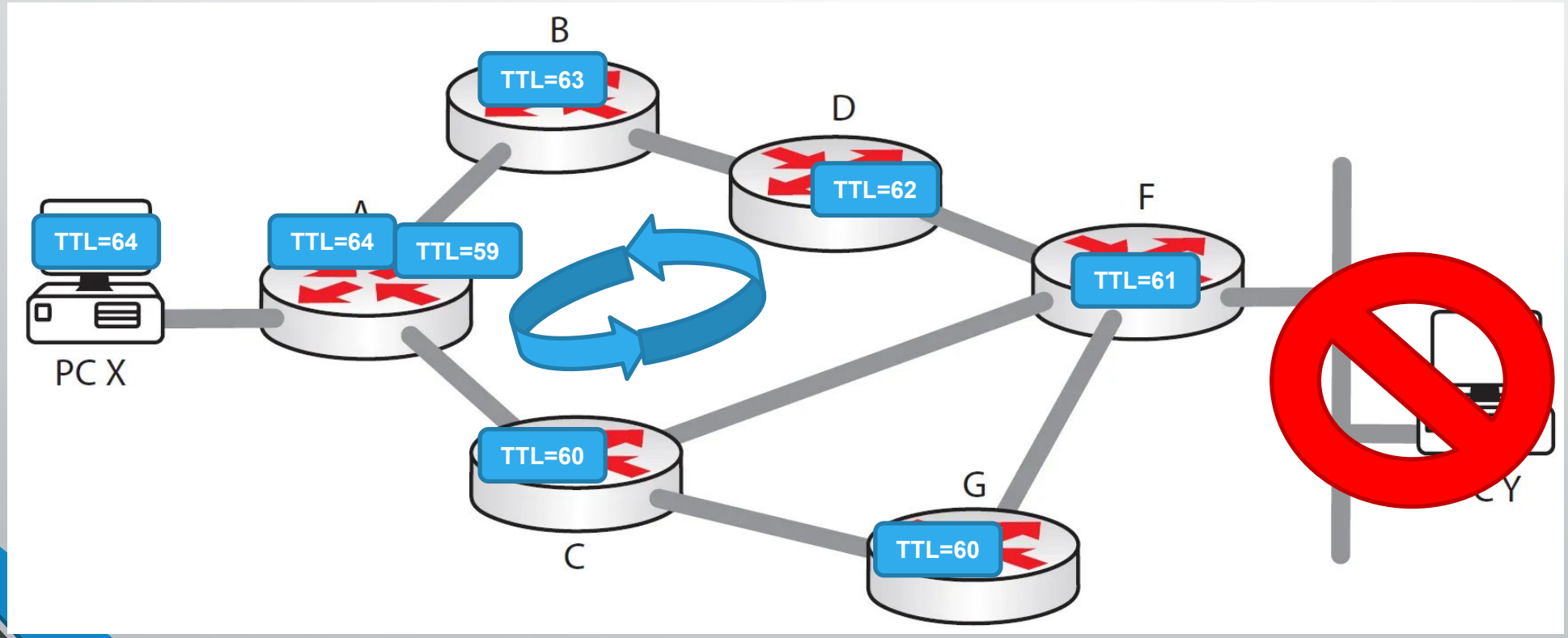
Time to Live - TTL

- Maximum number of **hops** (routers) a packet can traverse before being discarded.
- At each hop, the TTL is decreased by **1**.
- When the TTL reaches **0**, the packet is dropped.
- And an error message is sent back to the sender.

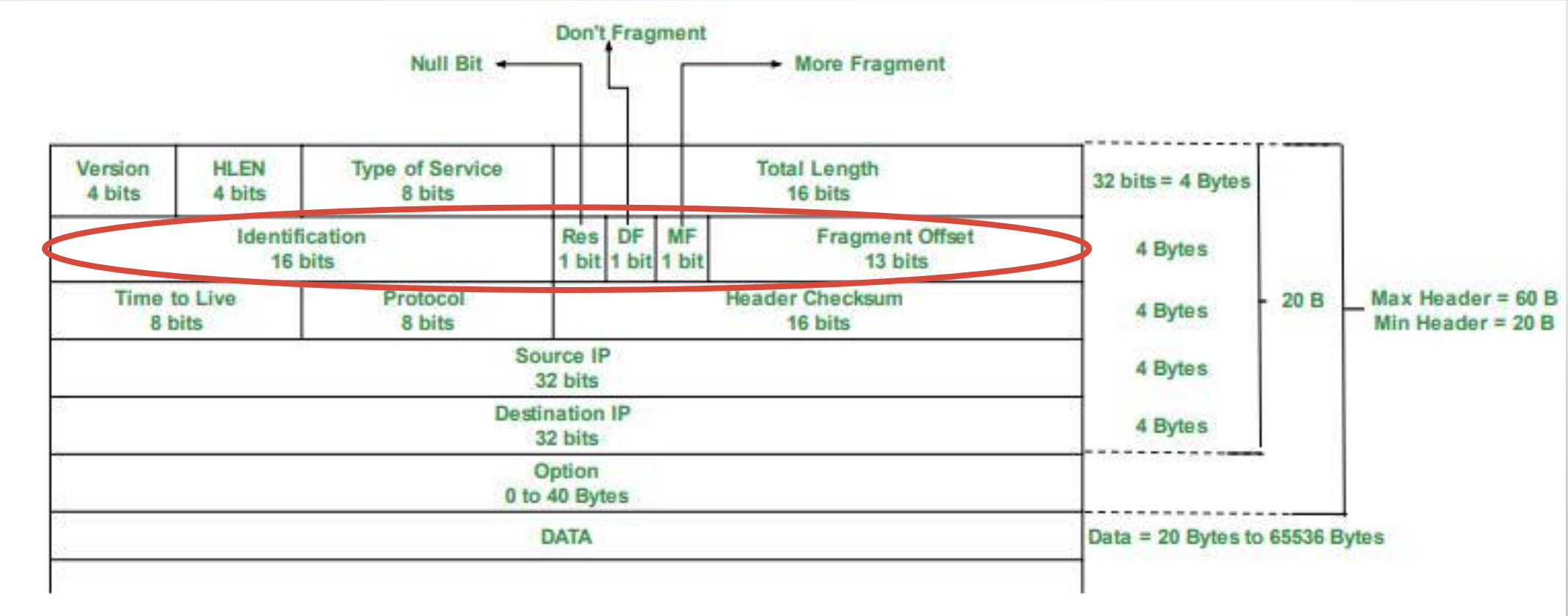


Time to Live - TTL

- **Not** just the "value of hops"
- It's a mechanism to prevent packets from **looping endlessly** in the network.
- Ensure finite packet lifetimes.

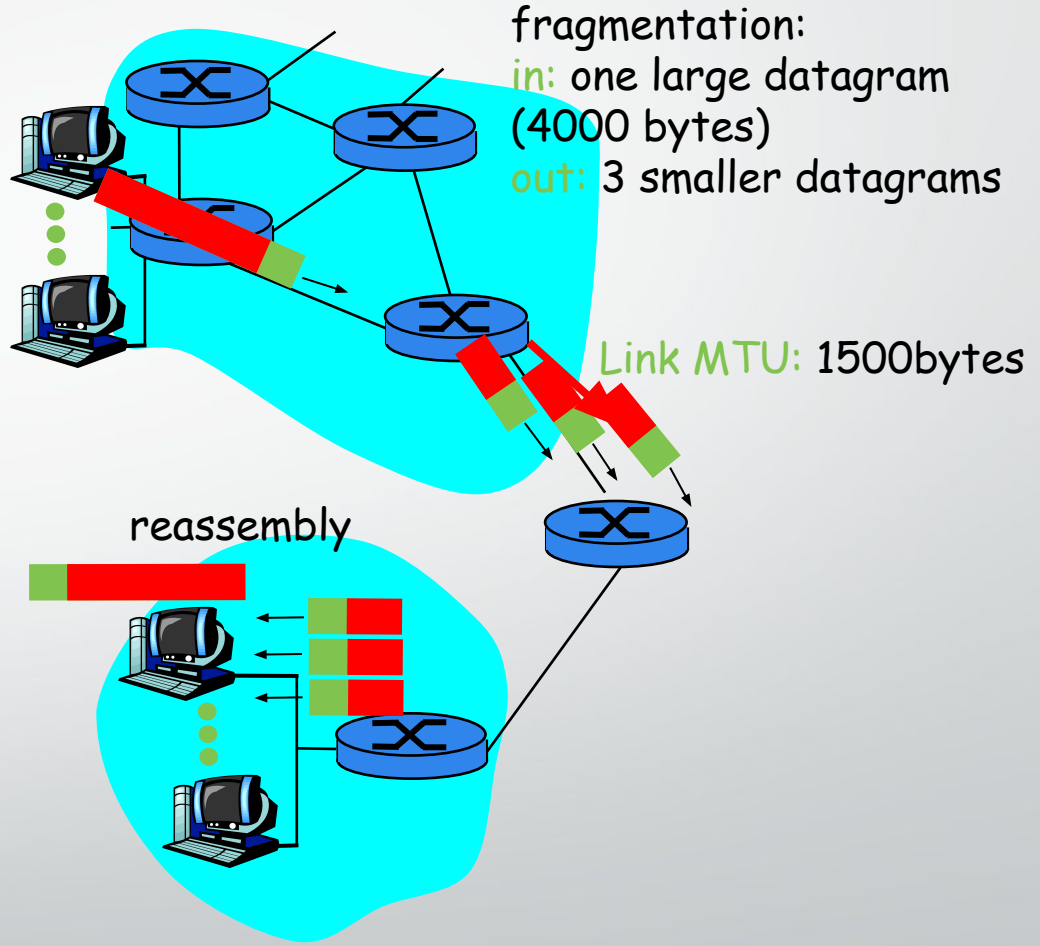


IPv4 Datagram Format



IP Fragmentation & Reassembly

- Network links have **MTU**
  - Maximum transmission unit or maximum transfer size
  - Different link types have different MTUs
  - Fragments reassembled at the destination



IP Fragmentation & Reassembly

Original IP Datagram

Identifier	Total Length	DF May / Don't	MF Last / More	Fragment Offset
345	5140	0	0	0

Data 5140=> 20(H)+5120(D)

MTU=1500 => 20(H)+1480(D)

1st fragment : 5120-1480=3640

2nd fragment : 3640-1480=2160

3rd fragment : 2160-1480=680

Last/4th Segment : 680+20=700

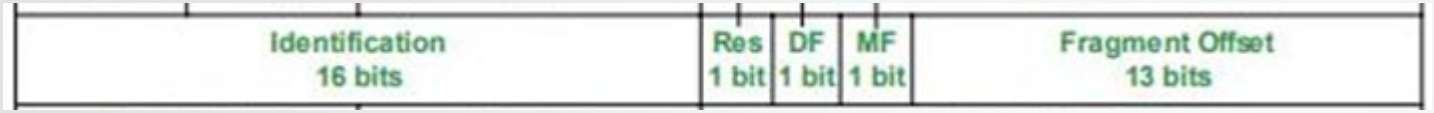
IP Fragments (Ethernet)

Identifier	Total Length	DF May / Don't	MF Last / More	Fragment Offset
345	1500	0	1	0
345	1500	0	1	185
345	1500	0	1	370
345	700	0	0	555

1st / start byte number

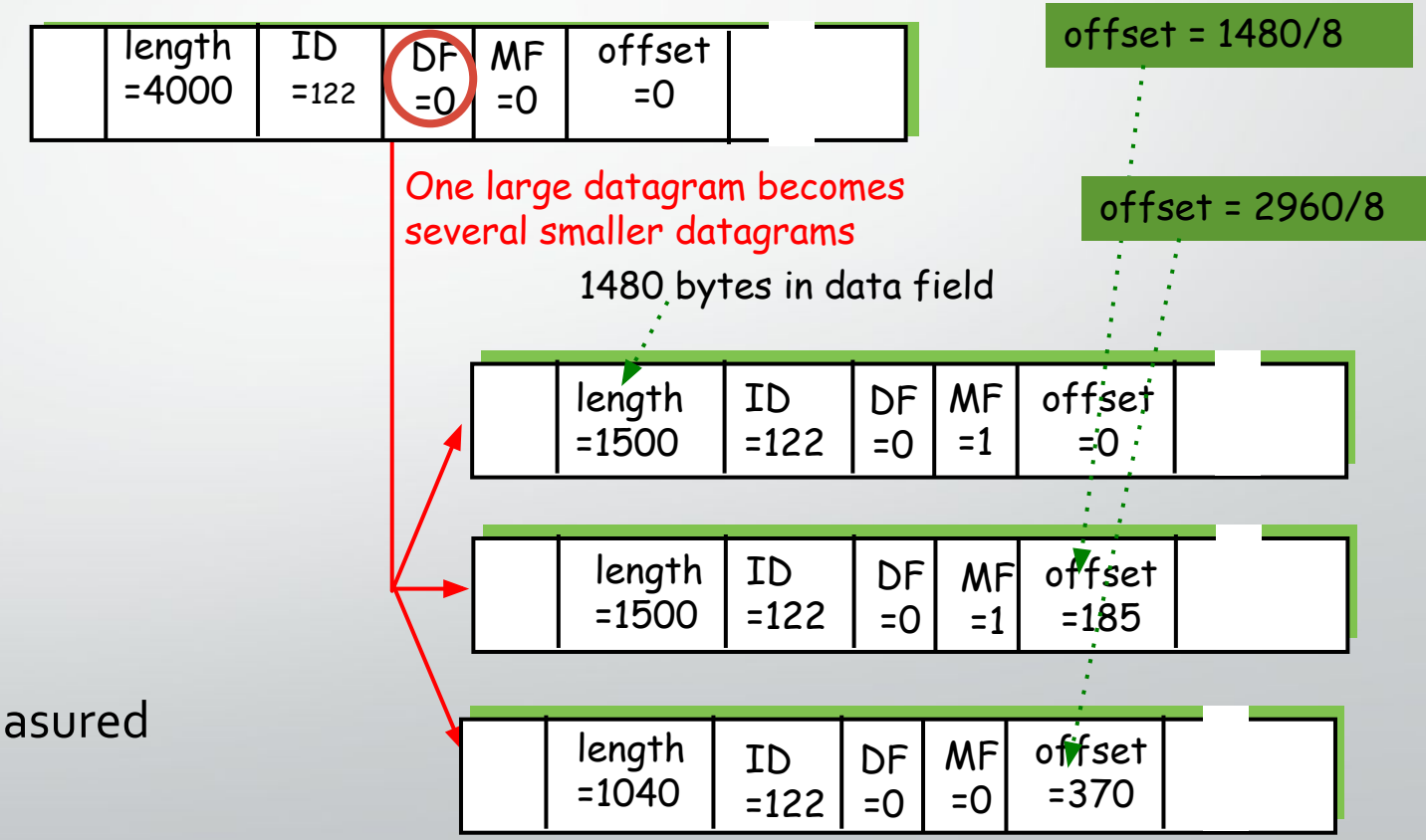
Data Bytes	Fragment Offset
0 -1479	0/8=0
1480-2959	1480/8=185
2960-4439	2960/8=370
4440-5119	4440/8=555

IP Fragmentation & Reassembly



Extra Example

- Example:
  - 4000 Bytes of datagram
  - MTU = 1500 Bytes
- **DF – Don't Fragment Bit**
  - Value 0 or 1
- **Fragment Offset**
  - The value of the offset is measured in units of 8 bytes.



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ICMP

Internet Control Message Protocol

ICMP

- It helps devices send error messages and status updates.
- **Functions**:
  - Reporting errors in the network
  - Checking reachability (Is the host alive?)
  - Diagnosing delays and congestion



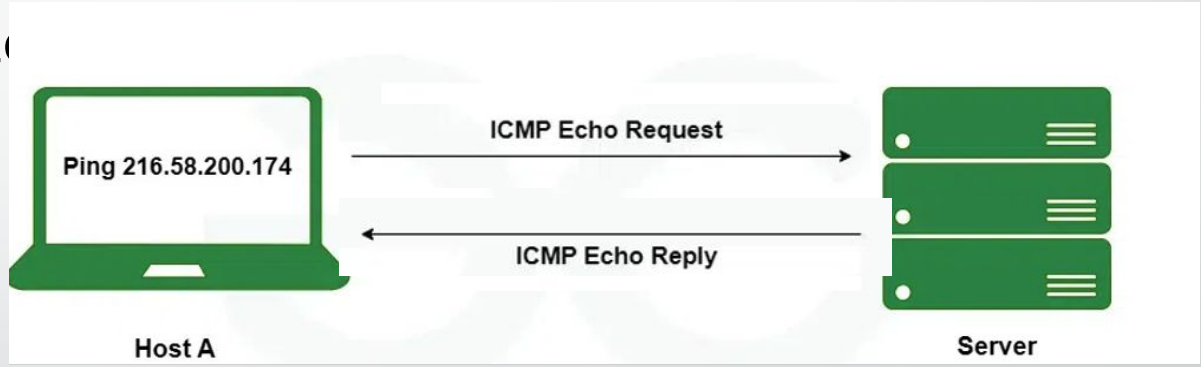




- **Key Points**:
  - ICMP does not carry actual user data
  - Mainly used by the **operating systems** for network management
    - **Example of ICMP in practice**
      - **Ping**
      - **Traceroute**

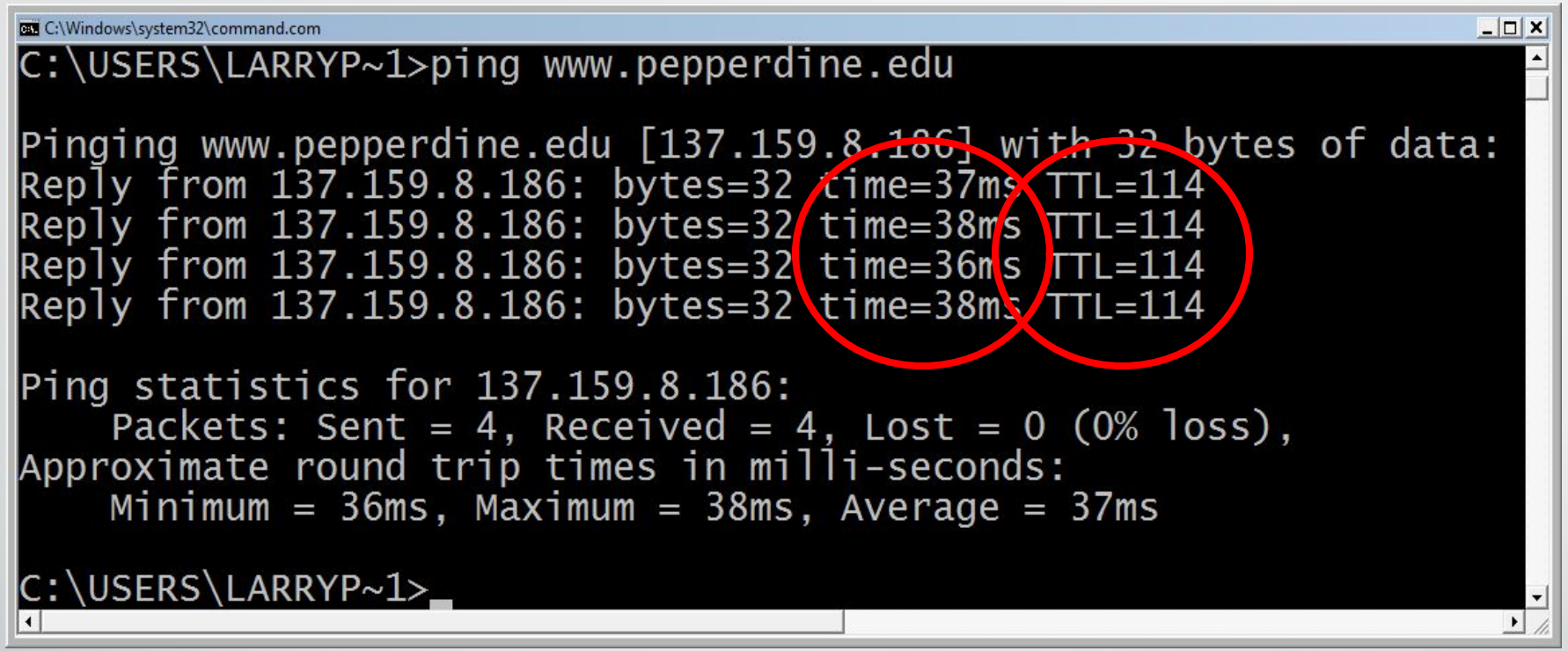
Ping

- Packet Internet Groper and is a network utility tool.
- **Purpose**:
- Checks if a device is reachable
- Measures the time it takes for data t
- **Mechanism**:
- Sends ICMP Echo Request packets
- Receives Echo Reply packets.
- **Results**:



- Commands:
  ping 216.58.200.174

Ping

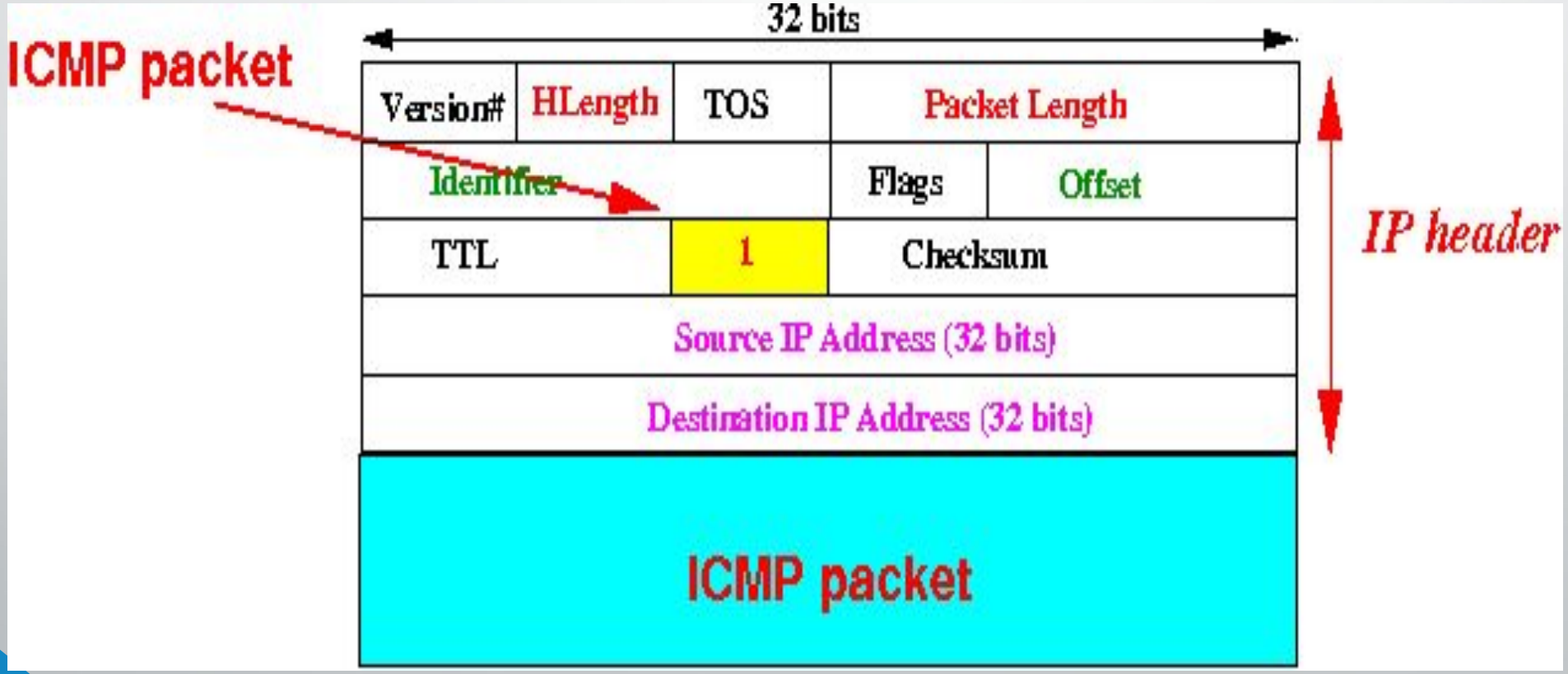


- **Questions :**
- Why 4 replies?
- What the time refer to?

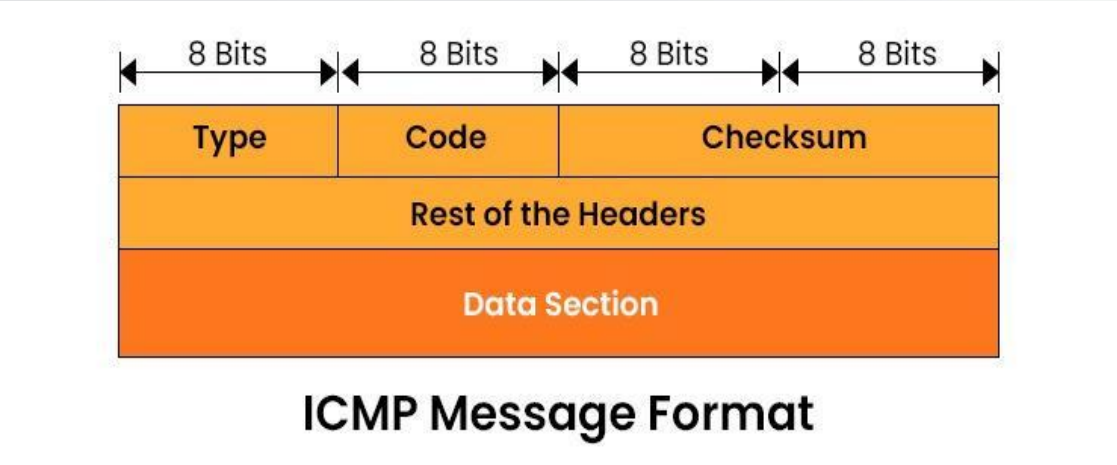
ICMP Packet Format

IP Header 20 bytes	IP Payload/ Data

ICMP Header 8 bytes	ICMP Data (variable)



ICMP Packet Format



ICMP Message Format

ICMP Type	ICMP Code	Description
0	0	Echo Reply (used by ping)
3	0	Destination Network Unreachable
3	1	Destination Host Unreachable
3	3	Destination Port Unreachable
8	0	Echo Request (used by ping)
11	0	TTL Expired (used by traceroute)

Type	Code	Description
0 – Echo Reply	0	Echo reply
3 – Destination Unreachable	0	Destination network unreachable
	1	Destination host unreachable
	2	Destination protocol unreachable
	3	Destination port unreachable
	4	Fragmentation needed and DF flag set
	5	Source route failed
5 – Redirect Message	0	Redirect datagram for the Network
	1	Redirect datagram for the host
	2	Redirect datagram for the Type of Service and Network
	3	Redirect datagram for the Service and Host
8 – Echo Request	0	Echo request
9 – Router Advertisement	0	Use to discover the addresses of operational routers
10 – Router Solicitation	0	
11 – Time Exceeded	0	Time to live exceeded in transit
	1	Fragment reassembly time exceeded
12 – Parameter Problem	0	Pointer indicates error
	1	Missing required option
	2	Bad length
13 – Timestamp	0	Used for time synchronization
14 – Timestamp Reply	0	Reply to Timestamp message

Unsuccessful Ping

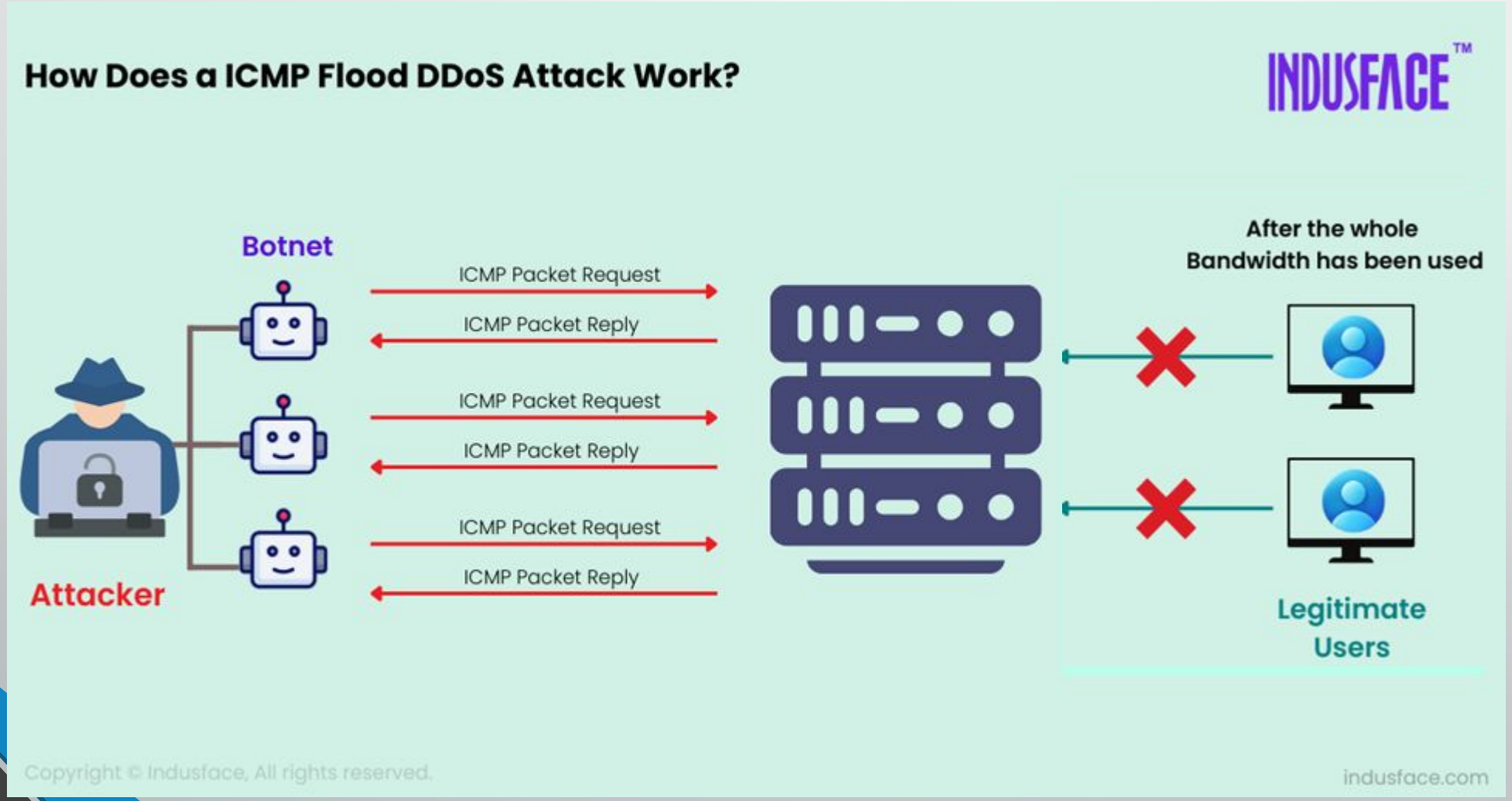
```
C:\>ping 10.2.104.2

Pinging 10.2.104.2 with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 10.2.104.2:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

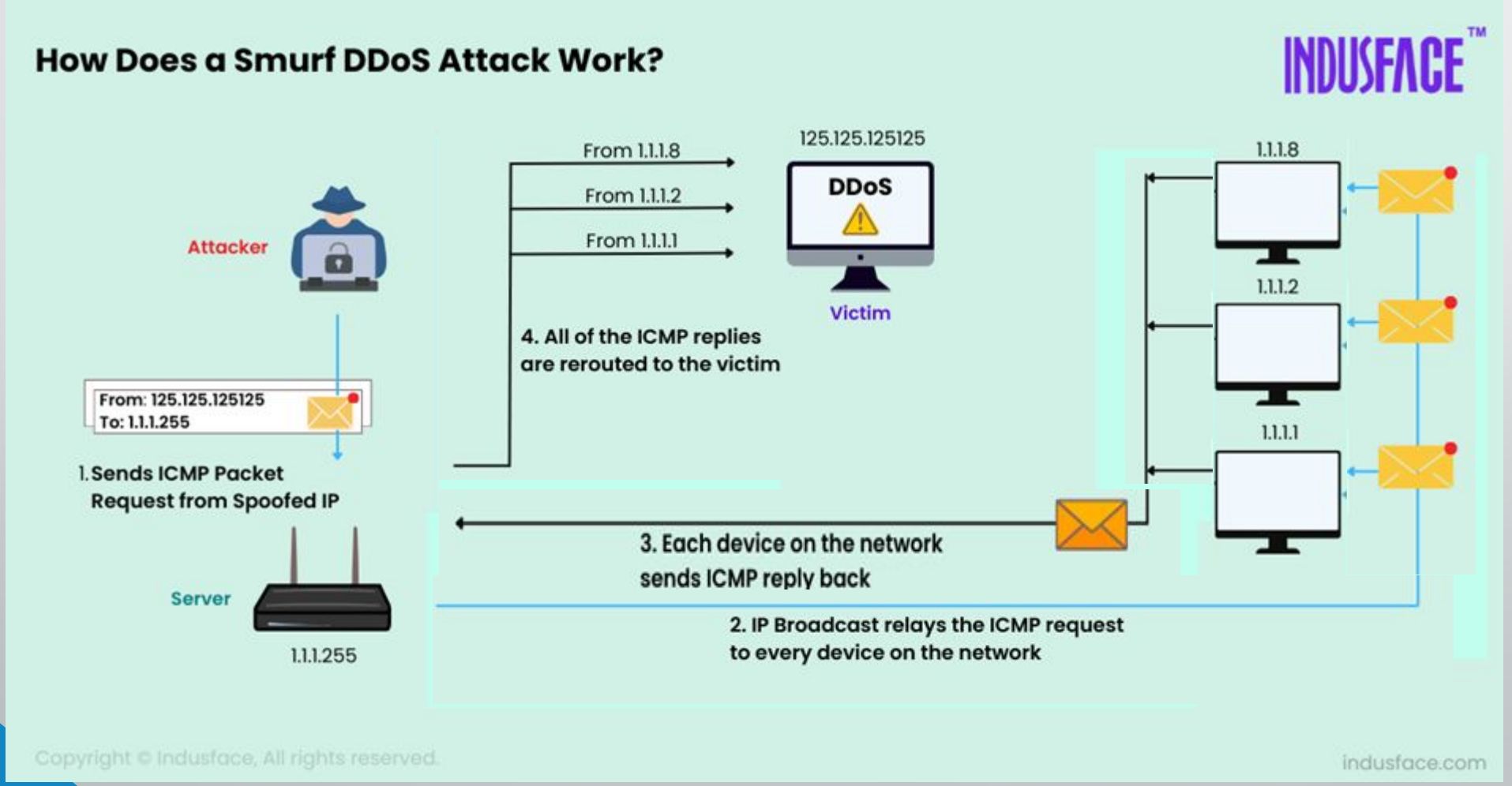
Ping Attacks

- **ICMP DDOS attack – Zombie Attack:**

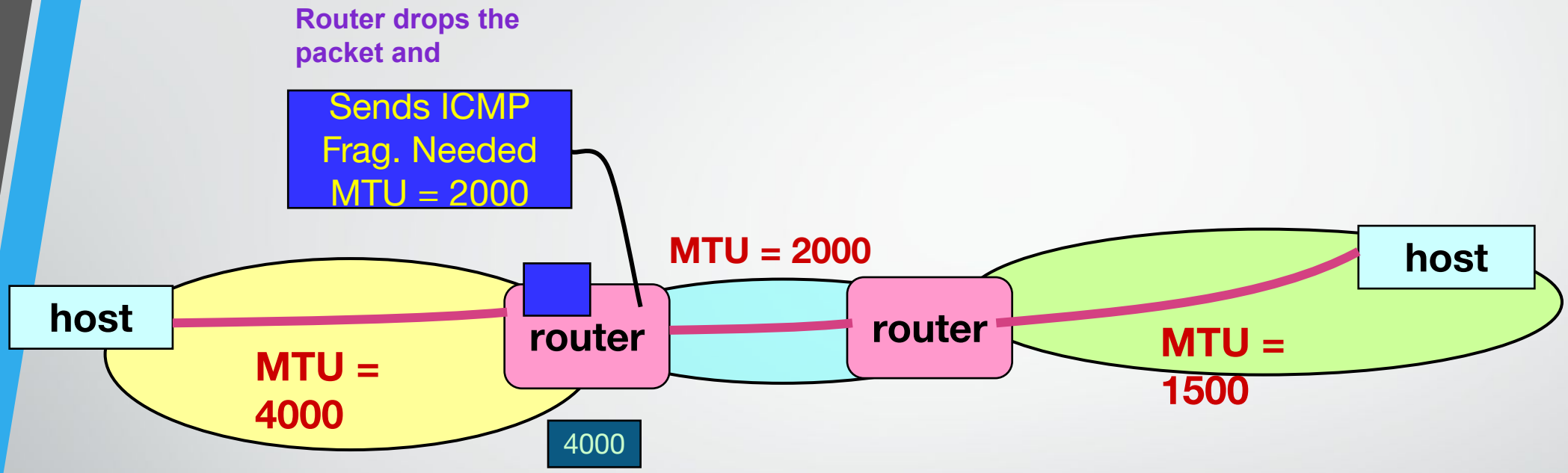


Ping Attacks

- **ICMP DDOS attack – Packet magnification (or ICMP Smurf):**



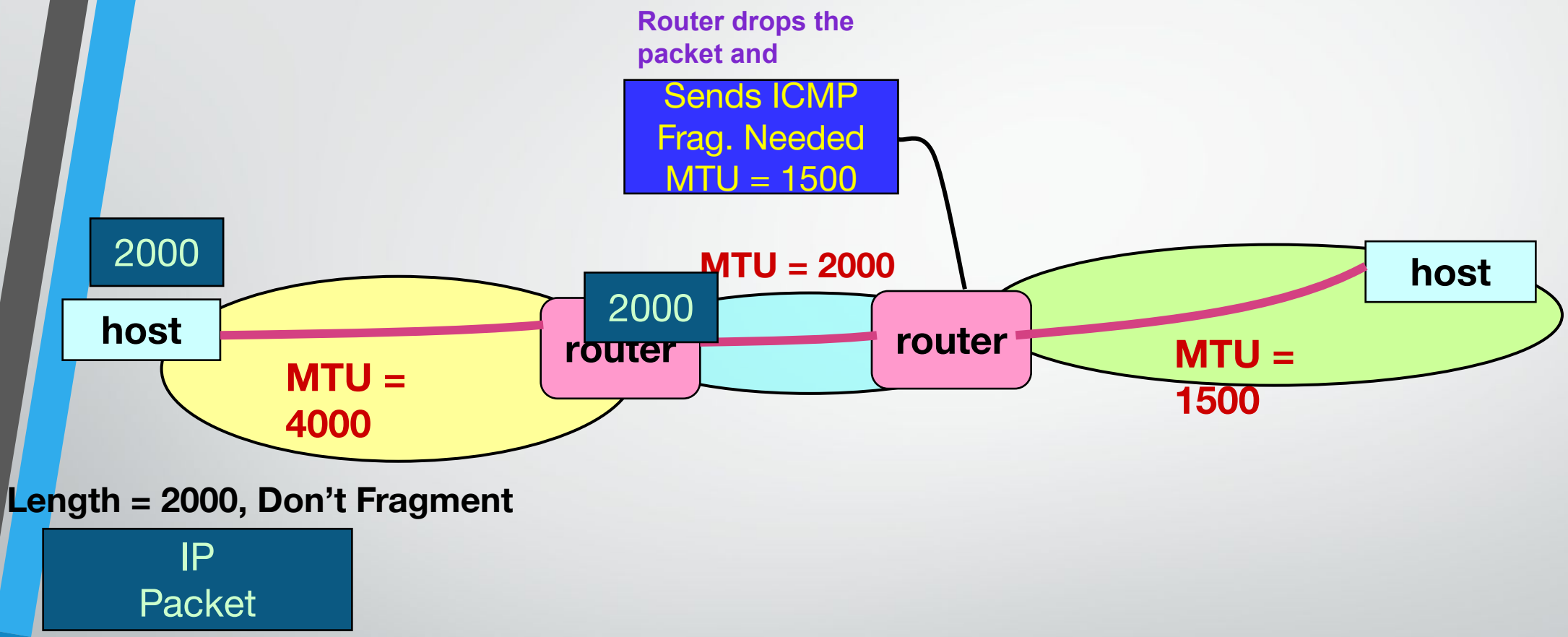
IP MTU Discovery with ICMP



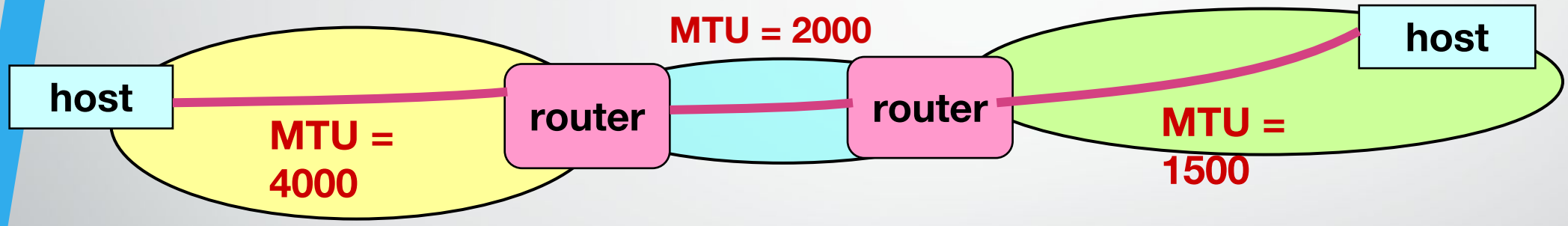
Length = 4000, Don't Fragment

IP Packet

IP MTU Discovery with ICMP



IP MTU Discovery with ICMP



Length = 1500, Don't Fragment

IP Packet

When successful, no reply at IP level

"No news is good news"

Higher level protocol might have some form of acknowledgement

Traceroute

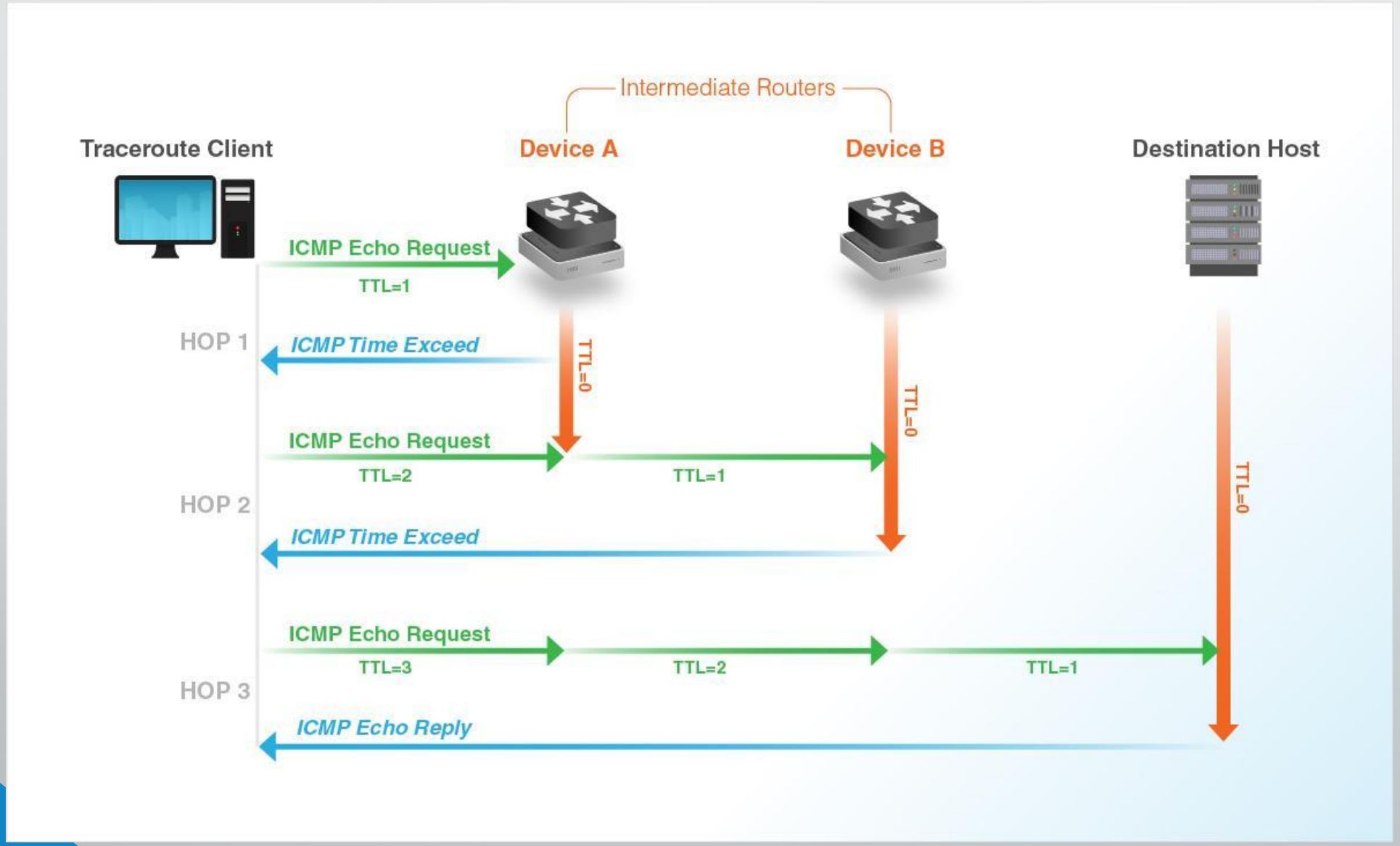
- Shows the path your packets take through different routers to reach a destination.

- **Purpose**:
  - Identifies the routers or hops data passes through to reach its destination mostly for troubleshooting

- **Mechanism**:
  - Sends ICMP packets with TTL = 1, 2, 3…
  - Each router where TTL becomes o sends ICMP Time Exceeded
  - Traceroute uses these replies to list each hop in order



Traceroute


Intermediate Routers
Traceroute Client
Device A
Device B
Destination Host
ICMP Echo Request
TTL=1
HOP 1
ICMP Time Exceed
TTL=0
ICMP Echo Request
TTL=2
TTL=1
TTL=0
HOP 2
ICMP Time Exceed
ICMP Echo Request
TTL=3
TTL=2
TTL=1
TTL=0
HOP 3
ICMP Echo Reply

Traceroute

- **Results** The IP or hostname of every hop and the time each hop takes (latency)

```
Microsoft Windows [Version 10.0.22000.613]
(c) Microsoft Corporation. All rights reserved.

C:\Windows\system32>tracert www.google.com

Tracing route to forcesafesearch.google.com [216.239.38.120]
over a maximum of 30 hops:

  1    <1 ms    <1 ms    <1 ms  192.168.0.1
  2     2 ms     2 ms     2 ms
  3     5 ms     5 ms     7 ms  172.25.0.137
  4     5 ms     5 ms     5 ms  172.16.2.158
  5    18 ms    17 ms    17 ms  72.14.216.48
  6    18 ms    17 ms    17 ms  108.170.240.225
  7    17 ms    16 ms    17 ms  142.251.52.49
  8    18 ms    18 ms    18 ms  any-in-2678.1e100.net [216.239.38.120]

Trace complete.
```

- Commands:
  - Unix: **traceroute**
  - Cisco IOS: **traceroute (trace)**
  - DOS: **tracert**

Command Prompt

```
Microsoft Windows [Version 10.0.19045.5131]
(c) Microsoft Corporation. All rights reserved.

C:\Users\skazi>traceroute www.yahoo.com
'traceroute' is not recognized as an internal or external command,
operable program or batch file.

C:\Users\skazi>tracert www.yahoo.com

Tracing route to me-ycpi-cf-www.g06.yahoodns.net [27.123.42.205]
over a maximum of 30 hops:

  1     4 ms     1 ms     1 ms  172.18.192.1
  2     *        *        *     Request timed out.
  3     3 ms     1 ms     1 ms  172.31.2.129
  4     1 ms     1 ms     1 ms  10.151.6.89
  5     1 ms     2 ms     1 ms  10.0.100.5
  6     2 ms     1 ms     1 ms  202.4.100.253
  7     1 ms     2 ms     2 ms  GI0-2-2-aggr01.as58656.net [103.12.177.1]
  8     2 ms     2 ms     2 ms  10.12.176.237
  9     3 ms     2 ms     2 ms  103.16.155.149
 10     2 ms     1 ms     1 ms  103.16.152.30
 11    11 ms    11 ms    10 ms  103.16.152.82
 12     *       51 ms    51 ms  103.16.153.21
 13    51 ms    51 ms    51 ms  103.16.153.18
 14    57 ms    57 ms    57 ms  ae6-1538.rt.eqx.sin.sg.retn.net [87.245.240.208]
 15    62 ms    63 ms    62 ms  ix-be-20.ecore4.esin4-singapore.as6453.net [180.87.54.66]
 16    64 ms    65 ms    64 ms  if-bundle-18-2.qcore2.esin4-singapore.as6453.net [180.87.108.80]
 17    70 ms    70 ms    71 ms  180.87.55.59
 18     *        *        *     Request timed out.
 19    69 ms    70 ms    78 ms  14.143.59.46.static-mumbai.vsnl.net.in [14.143.59.46]
 20    68 ms    68 ms    67 ms  e2-ha.ycpi.ina.yahoo.com [27.123.42.205]

Trace complete.

C:\Users\skazi>_
```

Using Tracert

Hop 1: Our local router or gateway (private IP address).
Hops 2–5: Internal routing within Bracu ISP's private network (non-public IPs).
Hop 6: First public IP, ISP's gateway to the internet.
Hops 7–9: Routing through regional and backbone ISPs.
Hops 10–13: Routing through Singapore (a major internet hub).
Hops 14–19: Routing through Indian networks, ending in Mumbai.
Hop 20: Final destination—Yahoo's server, located in India, near Mumbai.

Traceroute: Another example

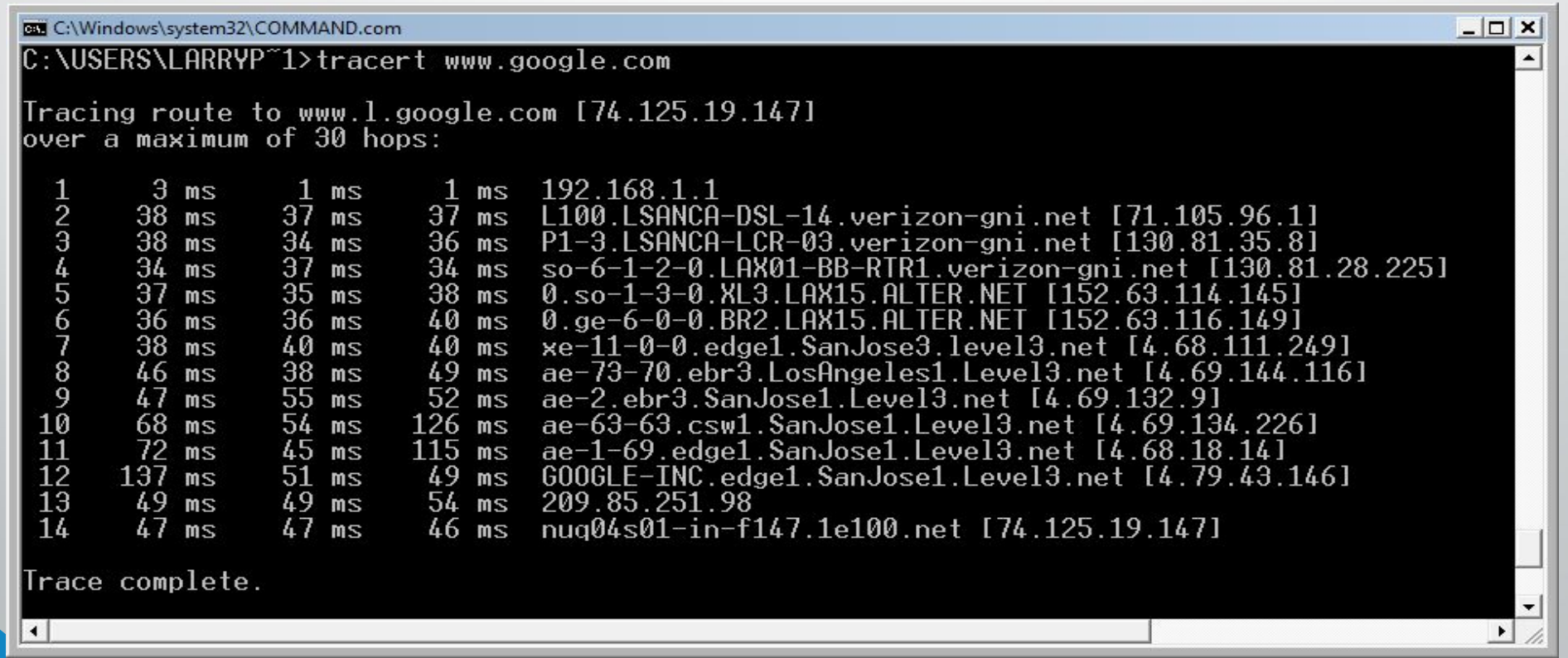
Hop 1: User LAN router

Hops 2-4: Verizon network (a backbone ISP)

Hops 5-6: Alternet (a backbone ISP)

Hops 7-11: Level 3 (a backbone ISP)

Hops 12-14: the Google LAN



Traceroute: Request Timed Out

This message indicates that the router security settings keep it from revealing its identity or the router and connection are slow.

```
*     *     *     Request timed out.
```

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The End